Jay Alexander Moore

Email: j.alex.moore@gmail.com
linkedin.com/in/jay-a-moore

I'm a kind and optimistic researcher driven by curiosity, big questions, and a passion for discovery through data. My work has focused on aggregating data and applying a wide range of analytical techniques to understand biological processes. I'm improving my scientific communication skills and exploring questions beyond my own field.

Next, I want to expand my use of agentic workflows and biological foundation models and contribute to their development.

Education

UK Dementia Research Institute at Imperial College London — London, UK
PhD, Clinical Medicine Research (Brain Sciences) — *May 2026 – Present*

- Thesis: *Mapping the Cellular Basis of Human Genetic Risk Through AI-curated Multiomics*

Institute of Child Health, University College London — London, UK
MSc, Personalised Medicine and Novel Therapies; Distinction — *Sep 2024 – Sep 2025*

- Dissertation: *Assays for Developing AAV.NDP Gene Therapy for the Treatment of Norrie Disease.* Used blood vessel organoids and ELISA to validate computational target identification.

Colby College — Waterville, ME, USA
BA, Computational Biology — *2014 – 2018*

Experience

UK Dementia Research Institute at Imperial College London — London, UK
PhD Student, Neurogenomics Lab (Nathan Skene) — *May 2026 – Present*
Research Assistant (Contract) — *Nov 2025 – Apr 2026*

- **Cell-Typing Genetic Risk in Neurodegeneration**: Develop agent-assisted curation workflows and cell-type expression specificity references for MAGMA.Celltyping and LDSC analyses. Source catalogues include scBaseCount (500M cells) and OpenGWAS (50K studies).
- **Parkinson's Neuropathology Meta-analysis**: Showed that substantial neuronal loss in Parkinson's disease extends beyond dopaminergic and pigmented neurons, challenging explanations centred on dopamine or neuromelanin. Co-first author of the bioRxiv preprint (in submission).
- **UK DRI Agentic AI Fleet**: Lead implementation of Hermes Fleet for **12 UK DRI labs** with Mathieu Bourdenx and Nathan Skene. Develop knowledge graph and retrieval-augmented generation workflows to connect research documents and data for hypothesis generation and research planning.
- **Anthropic AI for Science**: Was accepted to Anthropic's AI for Science Program with **US$50,000 in token credits** for Claude Science and API use, supporting six months of agentic research.

Curileum Discovery Ltd — London, UK
Biomedical Scientist — *Oct 2022 – Mar 2025*

- **Drug Discovery at a Startup**: Led gene and pathway enrichment analyses of bulk mRNA-seq from tissue and organoid models, supporting preclinical development of a compound to prevent gut polyps and a cell therapy for fistulas.
- **Wet Lab**: Completed training in primary cell culture, organoid differentiation protocols, qPCR, and flow cytometry.

Foundation Medicine Inc. — Cambridge, MA, USA
Scientist I, Cancer Genomics Research — *Sep 2021 – Jul 2022*
Senior Research Associate, Cancer Genomics Research — *Dec 2019 – Sep 2021*
Research Associate, Cancer Genomics Research — *Jul 2018 – Dec 2019*

- **HRDsig Development**: Developed **HRDsig**, a cancer genomics algorithm now used on **thousands of clinical samples each week** at Foundation Medicine to identify homologous recombination deficiency. Named inventor on two patent applications; first author of the *JCO Precision Oncology* paper (2023) and second author of the pancreatic cancer follow-up in *The Oncologist* (2023).
- **Real-World Cancer Genomics**: Co-authored **11 cancer-genomics papers**, including two in *Cancer Discovery* on HLA loss and immune evasion (2021) and small cell lung cancer subtypes (2023). Delivered genomic analyses for Roche/Genentech R&D and clinical researchers.

Earlier Research Experience

- **2017-2018 - Research Assistant, Colby College.** Analysed insect microbiome and transcriptome data; founded Colby Genomics Club, a student group that operated an Oxford Nanopore MinION sequencer.
- **Summer 2017 - Cancer Genomics Research Intern, Foundation Medicine.** Supported machine-learning cell-of-origin prediction; published in *Future Oncology* (2021).

Scientific Community, Teaching and Leadership

Teaching and Workshops

- **Agentic AI for Neuroscience:** Organised seminars hosted by UK DRI and led practical workshops with speakers from Tessl, StackOne, ClawBio, Edison Scientific, and Kiin Bio (April–June 2026). Created a public tutorial and YouTube playlist.
- **Festival of Genomics and Biodata:** Will lead *Agentic AI for Life Sciences: Build Your Own AI Research Assistant* (27 January 2027).

Events and Community

- **ClawBio:** Joined Manuel Corpas one week after ClawBio was formed to develop reusable AI skills for biology. One week later, co-hosted the Imperial ClawBio Hackathon for **100+ attendees** and delivered a hands-on workshop (March 2026). Support hackathons, talks, and the public events calendar. ClawBio was featured in *Nature* (2026).
- **Future of Biology Salons:** Co-organise events and helped secure **£10,000** from Imperial I-X's AI-in-Science and Lab Automation Event Fund for four events, including an Imperial demo day with Transfyr and an Oxford Tech Week salon with Unfabricated.
- **Wider Scientific Community:** Regularly attend events focused on research, lab automation, biotechnology, and longevity, including Society of Tech Advancement, The Coordination Problem, 50Y, future.bio, London Longevity, and Deep Tech London.

Selected Publications

- Challenging selective vulnerability in Parkinson's disease: a systematic review and meta-analysis. *bioRxiv* (2026). **Co-first author**; preprint (in submission). DOI
- Genomic profiling of small intestine cancers from a real-world data set identifies subgroups with actionable alterations. *JCO Precision Oncology* (2024). DOI
- Poor prognosis among radiation-associated bladder cancer is defined by clinicogenomic features. *Cancer Research Communications* (2024). DOI
- Pan-cancer analysis of copy-number features identifies recurrent signatures and a homologous recombination deficiency biomarker to predict poly (ADP-ribose) polymerase inhibitor response. *JCO Precision Oncology* (2023). **First author**. DOI
- A novel HRD signature is predictive of FOLFIRINOX benefit in metastatic pancreatic cancer. *The Oncologist* (2023). **Second author**. DOI
- Comprehensive genomic profiling and treatment patterns across ancestries in advanced prostate cancer: a large-scale retrospective analysis. *The Lancet Digital Health* (2023). DOI
- Integrative analysis of a large real-world cohort of small cell lung cancer identifies distinct genetic subtypes and insights into histologic transformation. *Cancer Discovery* (2023). DOI
- Somatic HLA class I loss is a widespread mechanism of immune evasion which refines the use of tumor mutational burden as a biomarker of checkpoint inhibitor response. *Cancer Discovery* (2021). DOI

Full publication list: Google Scholar.

Patent Applications

- *Methods and systems for identifying copy number signatures.*
  PCT/US2023/075910 2024
- *Systems and methods for classifying and treating homologous repair deficiency cancers.*
  US20230140123A1 2023